# Week 3: Statistical Modeling and Hypothesis Testing

*Data Science Internship Task Report*

Dataset: UCI/scikit-learn Wine recognition dataset; analysis uses the supplied cleaned_wine_dataset.csv. Observations: 178; chemical predictors: 13; target: wine_class with three classes.

## 1. Research objective and question

This task examines whether the 13 chemical measurements can predict the three-category wine_class label. A multinomial logistic regression classifier is fitted and evaluated on held-out data. An additional one-way ANOVA tests whether mean alcohol content differs across classes, providing an explicit statistical hypothesis test alongside predictive modeling.

## 2. Hypotheses

- ANOVA null hypothesis (H0): population mean alcohol content is equal for class_0, class_1, and class_2.
- ANOVA alternative hypothesis (H1): at least one class has a different population mean alcohol content.
- Modeling question: chemical predictors contain useful information for distinguishing the three wine classes. Predictive quality is assessed using held-out accuracy, macro-F1, confusion matrix, and stratified cross-validation.

## 3. Data preparation and assumptions

The cleaned dataset has 178 rows, 13 predictors, 3 target classes, 0 missing cells, and 0 duplicate rows. The target is categorical and predictors are numeric. A stratified 75/25 train-test split (random_state=42) preserves approximate class proportions. StandardScaler is included in a pipeline so means and standard deviations are learned only from training data, reducing data leakage.

## 4. Hypothesis test: one-way ANOVA

H0 and H1 are defined above. The one-way ANOVA result for alcohol is F = 135.078, p = 3.320e-36. At alpha = 0.05, reject H0. Mean alcohol differs for at least one class. Sample summaries are shown below; this test alone does not determine which pairwise differences are significant.

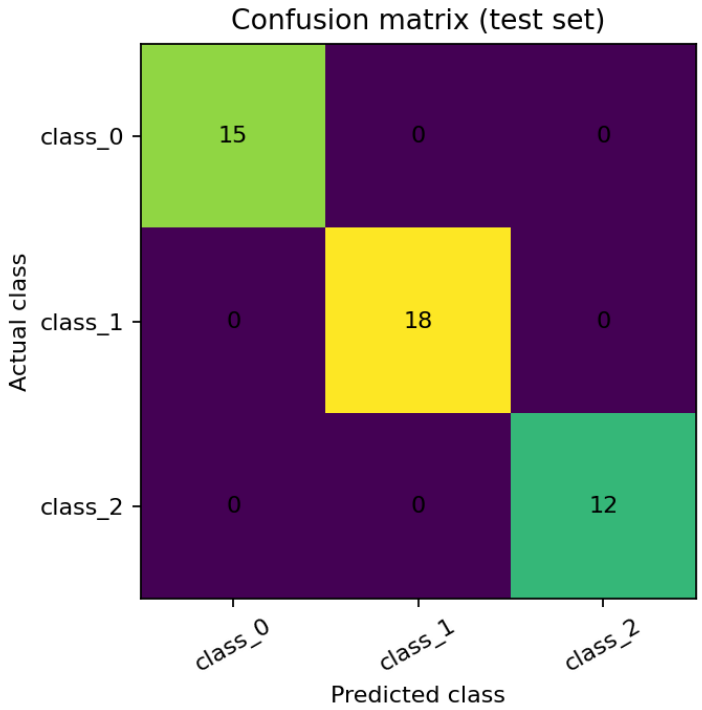
| Wine class | n | Mean alcohol | Std. dev. |
|---|---|---|---|
| class_0 | 59 | 13.745 | 0.462 |
| class_1 | 71 | 12.279 | 0.538 |
| class_2 | 48 | 13.154 | 0.530 |

ANOVA assumptions include independent observations, approximately normal residuals within groups, and comparable variances. A more complete inferential workflow would include Q-Q/residual plots, a variance-homogeneity test such as Levene's test, and post-hoc pairwise tests with multiplicity correction if warranted.

## 5. Multinomial logistic regression

A multinomial logistic regression model was trained on the 75% training subset using all 13 numeric chemical predictors. Standardization was performed inside a scikit-learn pipeline. The remaining 25% was held out until final evaluation. Logistic regression is suitable because wine_class has three nominal categories. Scaling is important because predictors have different units and magnitudes.

Test accuracy: 1.000 (45 correct predictions out of 45 test samples). Test macro-F1: 1.000.




The confusion matrix compares actual classes (rows) with predicted classes (columns). In this split, the classifier correctly classified every test sample. A perfect score on a small held-out set can occur by chance and should not be interpreted as guaranteed performance on future data.

## 6. Cross-validation

Five-fold stratified cross-validation with shuffling and random_state=42 produced fold accuracies of 0.972, 0.972, 0.972, 1.000, 1.000. Mean accuracy was 0.983 with fold standard deviation 0.015. The folds are broadly consistent, but the sample is small and cross-validation still uses the same benchmark dataset.

| Metric | Result |
|---|---|
| **Held-out test accuracy** | 1.000 |
| **Held-out macro-F1** | 1.000 |
| **5-fold mean accuracy** | 0.983 |
| **5-fold accuracy SD** | 0.015 |
| **ANOVA F-statistic** | 135.078 |
| **ANOVA p-value** | 3.320e-36 |

## 7. Diagnostics, interpretation, and limitations

- Independence: assumed from the dataset design; the notebook cannot verify how samples were collected.
- Logistic regression assumes an approximately linear relationship between each predictor and the class log-odds. Nonlinear patterns or interactions may not be fully represented.
- Multicollinearity among chemical predictors may make coefficients unstable. Correlation checks and variance-inflation diagnostics can be added if coefficient interpretation is a goal.
- The model report emphasizes prediction metrics; scikit-learn logistic regression does not directly provide the coefficient p-values reported by some inferential statistical models. No coefficient significance claims are made.
- No external validation dataset is available. High accuracy may reflect the clean, structured benchmark and should not be treated as evidence of equivalent performance in deployment.

## 8. Conclusion and reproducibility

The ANOVA provides strong evidence that average alcohol differs across wine classes. Multinomial logistic regression achieved perfect accuracy on this particular stratified test split and strong five-fold cross-validation accuracy. These results suggest the chemical measurements are informative for class separation in this dataset, while limitations around sample size, assumptions, and external validity remain. Run Week3_Statistical_Modeling_Hypothesis_Testing.ipynb with cleaned_wine_dataset.csv in the same directory. Required packages: pandas, numpy, matplotlib, scipy, scikit-learn, and Jupyter.

## 9. Dataset source

UCI Machine Learning Repository, Wine dataset; scikit-learn load_wine documentation. The analysis uses the supplied cleaned CSV and does not fetch new records.